

Chapter 1

Snow products

1.1 Study area

This thesis focuses on the Italian hydrological portion of the Greater Alpine Region (GAR), comprising all Italian territory located north of 43°N together with the Swiss portion of the Po basin. The main features of this territory are respectively:

- ▶ *the Italian Alps*, the southernmost part of the Alpine Arc. They extend for approximately 1200 km, covering an area of about $50\,000\text{ km}^2$. The highest peak, Monte Bianco, reaches 4810 m a.s.l., and summits above 3000 m a.s.l. are fairly common throughout the range. The considerable elevations sustain 901 glaciers, whose total surface area is roughly 368 km^2 (Smiraglia and Diolaiuti 2016).
- ▶ *the Po Plain*, a predominantly flat region shaped by an initial phase of marine sedimentation followed by later fluvial deposition. It is drained by the Po river, Italy's longest, which flows for approximately 652 km. Together with the Veneto-Friulian plains, it forms one of the country's principal agricultural and industrial areas, home to roughly 20 million people.
- ▶ *the Northern Apennines*, formed by the collision between the western margin of the African Promontory and the Sardinia-Corsica block. Their highest peak reaches only 2165 m a.s.l., insufficient to prevent the complete melt of the seasonal snowpack (Soldati and Marchetti 2018).

The complex orography of the study area, examined in greater detail in Figure 1.1, governs its climate. Indeed, the Italian Alps act as a barrier against cold air masses from the north, while the Apennines shield the Po Valley from the milder influence of the Tyrrhenian Sea. The main climate types found within in the study area are:

- ▶ *the Alpine climate*, dominant across the Alps and the Northern Apennines, associated with low temperatures both at night and during winter, and with a summer peak in precipitation.

- *the Po Valley climate*, characterized by cold winters and hot summers, with precipitation peaking in spring and autumn (Ministry for the Environment, Land and Sea 2007).

The high elevations of the Alpine Range, combined with the low winter temperatures, allow the Italian Alps to store substantial volumes of water in solid form, often exceeding 10 Gm^3 (Avanzi, Gabellani, Delogu, Silvestro, Pignone, et al. 2023). Some snowfall also occurs over the Norther Apennines, though in far smaller amounts. These differences in snow cover give rise to distinct hydrological regimes, as up to 50% of the annual stream flow of the main rivers originating in the Italian Alps consists of meltwater (Colombo et al. 2023), and peak discharge is reached in early summer when ablation is most intense. On the other hand, rivers draining the Northern Apennines rely predominantly on rainfall, and often experience low flow, or even run completely dry, during summer.

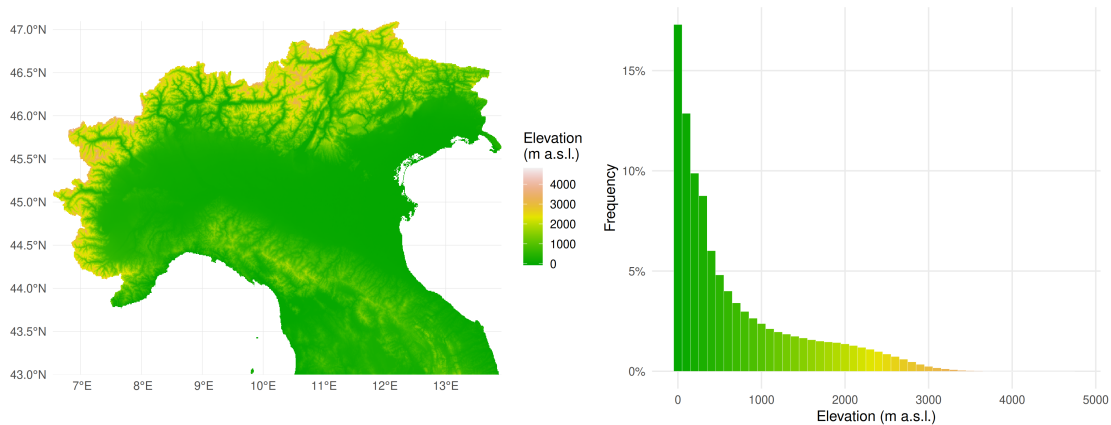


Figure 1.1: Topography and elevation distribution of the study area. The frequency histogram is based on 100 meters wide elevation bins. Elevation data were obtained from ALOS AW3D30 global digital elevation model (DEM).

1.2 MODIS

Thanks to its detailed spectral coverage of the Earth’s surface, NASA’s Moderate Resolution Imaging Spectroradiometer (MODIS) has become a key tool for mapping snow cover extent and duration across various regions, such as the Greater Alpine Region (Fugazza et al. 2021) or the Karakoram (Almagioni et al. 2025). One of the most recent studies employing MODIS focuses on the Italian territory (Manara et al. 2025), and therefore on the Italian Alps, which constitute the primary focus of this

thesis. This dataset, curated by the Department of Environmental Science and Policy of the University of Milan, provides daily snow cover information at the national scale and, in its most recent version, spans the period from 2000 to 2025. It consists of binary rasters, in which a value of 0 corresponds to bare ground, while a value of 1 denotes a pixel that is at least half covered by snow. The spatial resolution is approximately 500 m.

1.2.1 Dataset generation and gap-filling strategy

To obtain binary rasters describing the presence or absence of snow cover, Manara et al. 2025 proposed an operational processing chain designed to maximise prediction reliability. The procedure makes use of the MODIS Terra Snow Cover Daily L3 Global 500 m SIN Grid V061 (MOD10A1 V61) and MODIS Aqua Snow Cover Daily L3 Global 500 m SIN Grid V061 (MYD10A1 V61) products, both distributed by the US National Snow and Ice Data Center (NSIDC). In these datasets, an integer value is assigned to each grid cell according to the criteria summarised in Table 1.1.

Table 1.1: Integer values in NSIDC snow product.

Value	Condition
0	Snow free
1-100	Snow covered (NDSI value)
200	Missing data
201	No decision
211	Recorded during the night
237	Fresh water
239	Sea water
250	Cloud covered
254	Detector saturated
255	Fill value

Cloud cover can substantially reduce data availability across the study area, as the five year mean cloud cover can exceed 80 %. To address the resulting gaps in the MODIS record, Manara et al. 2025 first reduce the complexity of the dataset to three surface classes: snow covered, snow free and missing data. Integer values between 1 and 100 are reclassified as snow covered whenever the corresponding NDSI exceeded the critical threshold of 0.4, and as snow free otherwise. Pixels acquired at night, affected by cloud cover or flagged with an error condition are all labelled

as missing data, whereas freshwater pixels are reclassified as missing only when freshwater presence has been detected in less than 10 % of the available observations. Once the number of classes has been reduced to three, Manara et al. 2025 propose a gap filling procedure structured into six sequential phases, each exploiting different spatial and temporal information to progressively reclassify the missing pixels until the record is complete. These steps, adapted from the previous work of Gafurov and Bárdossy 2009 consist of the following:

- *Terra-Aqua combination*, which exploits the few hour offset between Terra and Aqua overpasses. Since Terra is considered more accurate than its counterpart, the snow map is built primarily from Terra values whenever snow presence or absence is known. If a Terra data is missing, the corresponding Aqua value is examined instead and retained if it is not itself missing. This step assumes that no snowmelt or snowfall occurs between the two overpasses.
- *Short range temporal filling*, which relies on combining snow cover information from adjacent dates to try filling the remaining gaps. For each grid point, the values recorded on the previous and following day are considered: if neither is missing and both agree on the pixel classification, the missing value is reclassified accordingly.
- *Elevation based filling*, which relies on the vertical extent of snow cover on a given day. For each day, the maximum snow line elevation H_s^{max} (above which every pixel is snow covered), and the minimum snow line elevation H_s^{min} (below which every pixel is snow free) are computed. A missing pixel located above H_s^{max} is thus reclassified as snow covered, while one located below H_s^{min} is labelled as snow free.
- *Direct neighbour spatial filling*, during which each missing pixel is compared with its four adjacent pixels. If at least three share the same class, that class is assigned to the missing pixel.
- *Elevation weighted spatial filling*, in which the 3×3 neighbourhood surrounding each missing grid point is examined. If any neighbouring pixel is snow covered and lies at a lower elevation than the central one, the missing pixel is reclassified as snow covered.
- *Pixel-wise temporal filling*, which consists of examining the subsequent values of a missing grid point until a snow covered or snow free classification is found. This value is then assigned to the missing pixel.

After the sixth step, the dataset is entirely free of missing values, providing snow cover information on a daily basis across the whole Italian peninsula.

1.2.2 Results

Following the approach adopted by the authors, particular attention was devoted to temporal snow metrics. In particular, the duration of the snow season (SCD) at a given location was computed as

$$\text{SCD}_{i,j} = \sum_{k=1}^n S_{i,j,k}, \quad (1.1)$$

thereby summing, over an hydrological year spanning from the 1st of October to the 30th of September, the binary values recorded for that specific pixel. In 1.1, k denotes the day of the hydrological year, while (i, j) represent the spatial coordinates of the pixel. The SCD is clearly related to the amount of accumulated snow, to temperature and to other climatic drivers. In order to assess whether the enviromental setting induces a shift in the onset or end of the snow season, a fixed date method is typically employed. The snow onset date (SOD) is defined as

$$\text{SOD}_{i,j} = \sum_{k=1}^D D - S_{i,j,k}, \quad (1.2)$$

where D corresponds to 116, namely the number of days between the beginning of the hydrological year and the 25th of January. Conversely, the snow end date (SED) can be computed as

$$\text{SED}_{i,j} = \sum_{k=D}^n D + S_{i,j,k}, \quad (1.3)$$

with D retaining the same meaning as in (1.2). These three metrics provide only cumulative information, thus disregarding transient snowfall and meltout events. In the following, only results regarding SCD will be presented: for the analogous analysis concering the onset and the end date of the snow season, the reader is referred to Appendix A.

As expected, MODIS proves capable of capturing the spatial inhomogeneities in snowpack lifetime across the Italian peninsula. This becomes particularly evident in the average SCD map, obtained as the mean of the annual SCD maps and shown in Figure 1.2. Indeed, mountainous regions are associated with the highest average SCD values, typically exceeding two hundred days of snow presence on the ground in high alpine catchments. The orography of the Italian Alps is closely reproduced, as valleys exhibit markedly lower snow cover duration than the surrounding peaks.

The Apennines remain snow free for substantially longer than the Alps, with only the most prominent massifs, such as Gran Sasso and Maiella, displaying snow behaviour somewhat comparable to their Alpine counterpart. On the other hand, the lowest SCD values are associated with lowland plains, where on average exceeding 5 days of snow on the ground is extrimately rare. SOD and SED exhibit a very similar behaviour to what was previously noted, as mountainous terrain strongly increases the likelihood of a prolonged snow season.

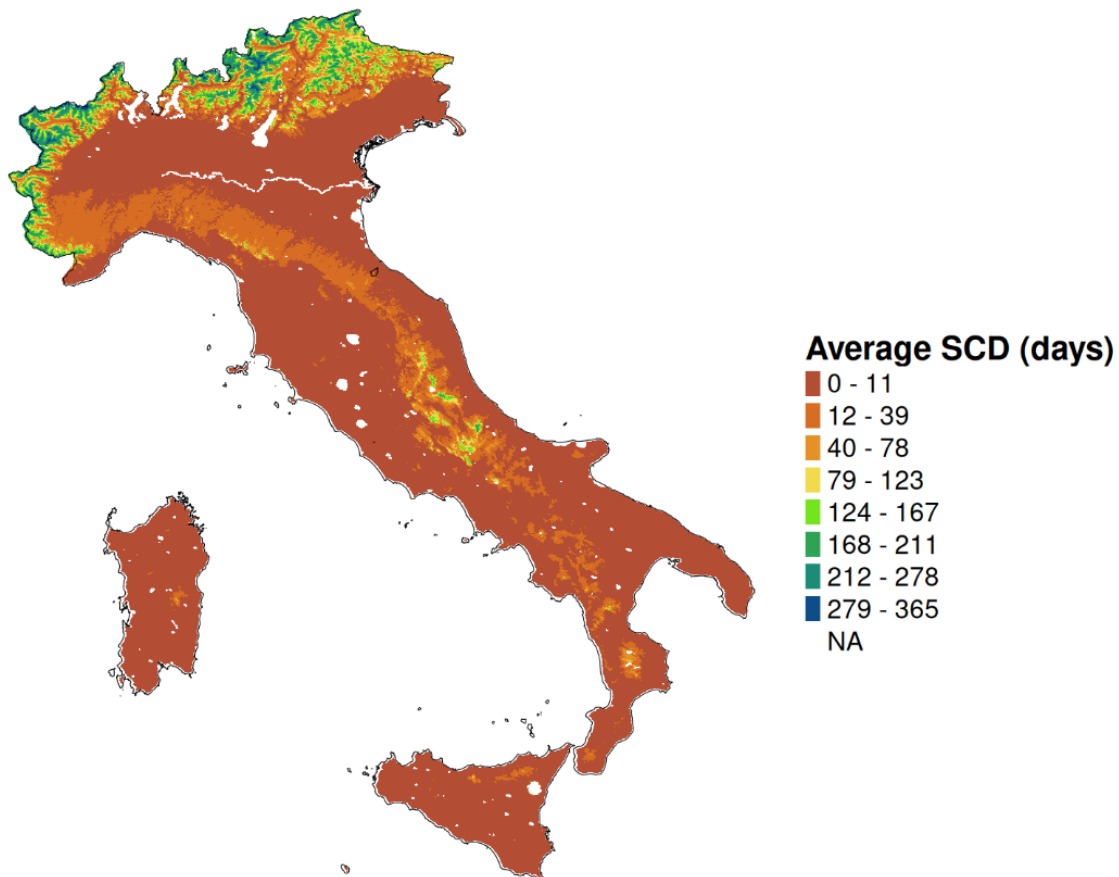


Figure 1.2: Average SCD map across the Italian peninsula for the 2001-2025 period. MODIS accurately reproduces the orography of the Italian Alps, as altitude strongly influence the persistence of snow on the ground. The lowest SCD values are associated with lowland plains, which remain snow free for most of the year.

The analysis of the average SCD map reveals that elevation plays a prominent role in determining both the spatial distribution and the temporal duration of snow cover. To gain deeper insights into this relationship, one possible approach is to divide the Italian territory into 5 m wide elevation bands, in order to plot the quantiles of the SCD distribution as a function of elevation. Notably, the trend shown in Figure 1.3 is clearly monotonic, since higher regions tend to be associated with longer

lasting snowpacks, even though a considerable spread is observed within each band. This wide range of possible SCD values indicates that elevation alone cannot fully account for snow distribution: a multilinear regression against several topographic and geographic variables shows that aspect and latitude also provide a non negligible contribution to snowpack behaviour (Manara et al. 2025).

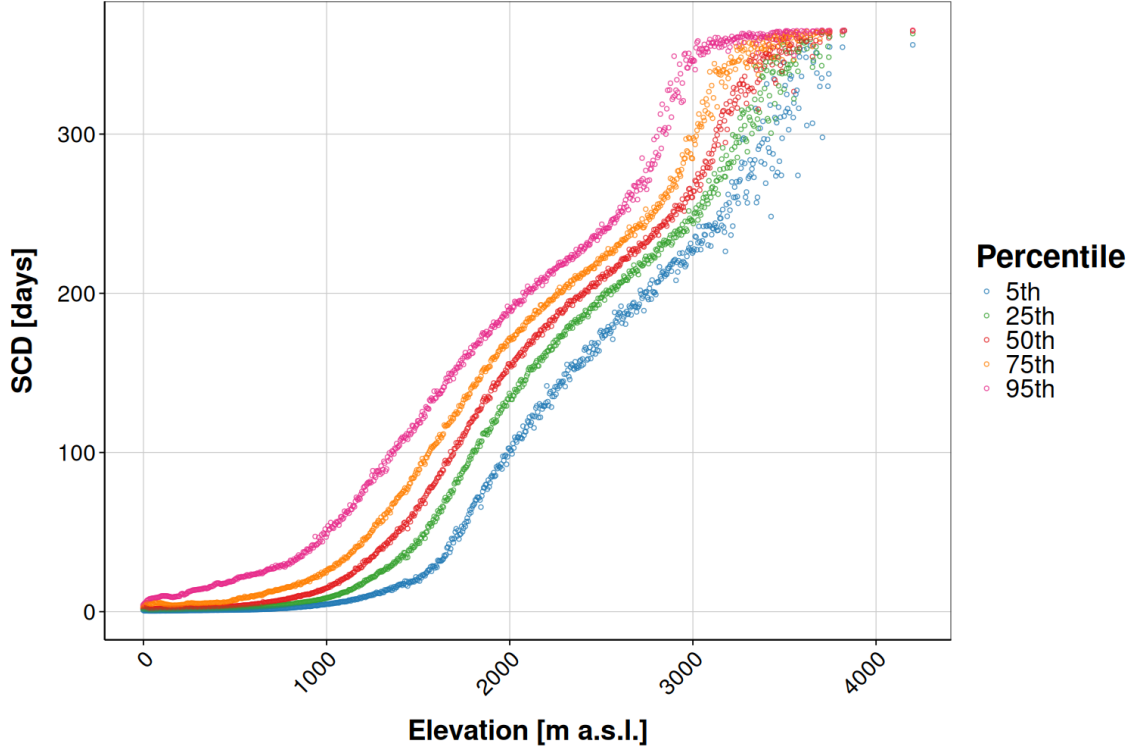


Figure 1.3: Quantiles of the average SCD distribution in each 5 m wide elevation band for the 2001-2025 period. The area taken into account is the whole Italian peninsula. A clear monotonic trend is visible.

The importance of aspect for the duration of the snow season is also evident when examining the spatial distribution of the difference between the SCD value of each individual pixel and the mean SCD per elevation band. In fact, in Figure 1.4 the Alpine valleys are clearly delineated, since south facing slopes experience, on average, a shorter snow season than north facing ones at the same elevation. The Easter Apennines can also be seen to be markedly snowier than the western side of the same ridge. This is attributed to cold continental air masses originating from the Balkan region, which become enriched with moisture as they cross the Adriatic Sea. This water vapor is subsequently released as snowfall through orographic forcing (Capozzi et al. 2025). This east to west gradient progressively weakens toward the Southern Apennines, which typically exhibit a shorter snow cover duration than the other Italian mountain ranges.

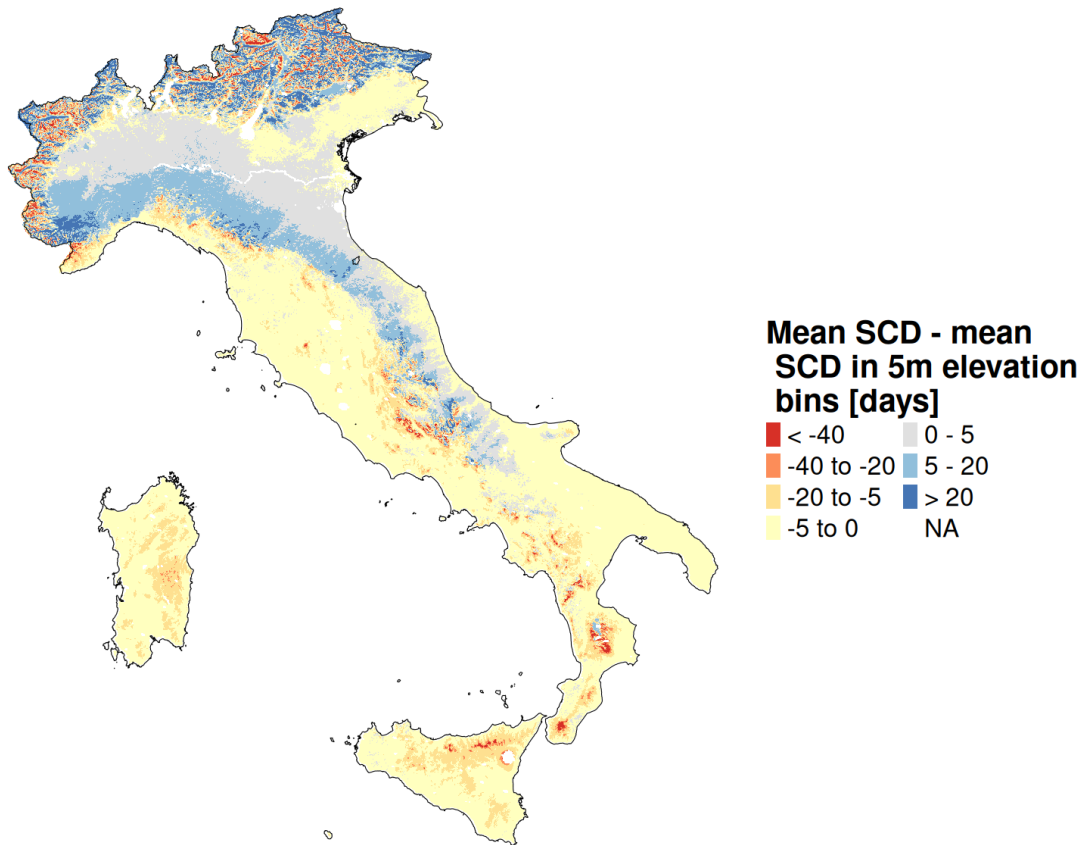


Figure 1.4: Spatial distribution of the difference between the SCD value of each individual pixel and the mean SCD per 5 m wide elevation band.

The snow cover extent and duration observed in any given year cannot, of course, be expected to closely match the 2001-2025 average, since the snowpack is highly sensitive to climatic drivers, such as winter precipitation and temperature patterns, that are themselves subject to considerable interannual variability. This translates into substantial year to year fluctuations in snow cover duration, which can be quantified through the pixel by pixel standard deviation from the mean, together with its ratio to the corresponding average SCD value. The spatial distribution of these two quantities is shown in Figure 1.5. The standard deviation exhibits a monotonically increasing relationship with up to approximately 2000 m a.s.l., beyond which it begins to decrease. This behaviour may be partially linked to glacier covered terrain, which, as previously noted, is treated by MODIS as snow covered throughout the whole year, even though bare ice conditions are commonly observed by the end of the melt season. The ratio displays a markedly different pattern, with lower values typically found in high alpine catchments, whereas the highest ones are generally observed over lowland plains. This trend follows directly from the considerably

longer SCD characterising mountainous terrain, which constraints the ratio regardless of the absolute variability.

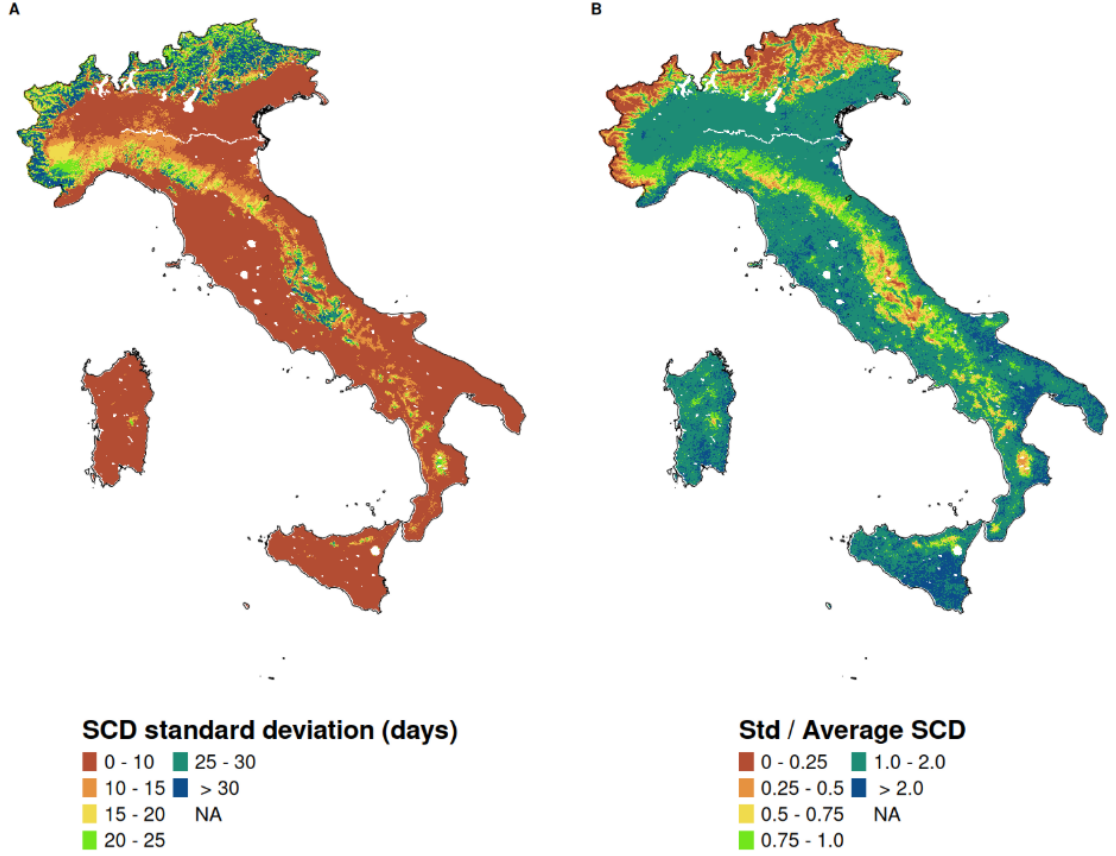


Figure 1.5: (a) Spatial distribution of SCD standard deviation from the mean, computed over the 2001-2025 period. (b) Spatial distribution of the ratio between the SCD standard deviation from the mean and the corresponding mean SCD value for the 2001-2025 period.

1.3 IT-Snow

IT-SNOW is a high resolution, multi-year snow reanalysis dataset covering the entire Italian peninsula at a spatial resolution of roughly 500 m. This dataset consist of daily maps of snow water equivalent (SWE), snow depth (HS), bulk snow density and liquid water content of the snowpack, spanning the period from September 1st, 2010 to August 31st, 2025 in its latest release. IT-SNOW is the output of S3M Italy, a real-time operational monitoring chain developed by CIMA Research Foundation and specifically designed for large scale applications, with the aim of serving as a valuable tool within flood forecasting frameworks. This model achieves a high level of performance by focusing on the key drivers of snow evolution, namely

total precipitation, air temperature, relative humidity, and total radiation, thereby providing a spatially consistent reconstruction of snowpack dynamics.

1.3.1 S3M

S3M, also known as *Snow Multidata Mapping and Modelling*, is a hydrology oriented cryospheric model developed to simulate the spatial and temporal evolution of seasonal snowpack. It adopts a raster based approach, in which neighbouring pixels are treated as independent units, so that horizontal mass and energy transport is neglected. All input and output variables are therefore spatially distributed, and the simulated snowpack represents the average conditions within each grid cell. The model provides daily (or even hourly) rasters of snow water equivalent (SWE), snow depth (HS), bulk snow density and liquid water content. From these outputs, the snow cover fraction (SCF) of each pixel can be computed as

$$\text{SCF} = \min \left(1, \frac{\text{SWE} [\text{mm}]}{0.1 \cdot \rho_s} \right), \quad (1.4)$$

where ρ_s is the bulk snow density (Kouki et al. 2023). Since a pixel is conventionally regarded as snow covered when at least half of its area is, binary snow extent maps are obtained by assigning a value of one to all grid points with $\text{SCF} > 0$, and zero elsewhere. Beside deepening the knowledge of snowpack properties, this quantity also provides a direct term of comparison with satellite snow cover products such as MODIS.

Within S3M, seasonal snow is partitioned into a dry component, corresponding to the ice matrix, and a wet component, given by the interstitial liquid water. These two constituents are modelled separately, and their mutual interactions govern the overall snowpack dynamics. The total snow water equivalent is thus obtained as

$$\text{SWE} = \text{SWE}_D + \text{SWE}_W, \quad (1.5)$$

so as the sum of its dry (SWE_D) and wet (SWE_W) contributions. The evolution of these two quantities is described by the following mass conservation equations:

$$\frac{d}{dt} (\text{SWE}_D) = S_f - M + R, \quad (1.6)$$

$$\frac{d}{dt} (\text{SWE}_W) = R_f + M - R - O, \quad (1.7)$$

where S_f and R_f denote the snowfall and rainfall mass fluxes, M and R the snowmelt

and refreezing rates and O the snowpack runoff mass flux. All the terms in equations (1.6) and (1.7) are expressed in mm w.e. h^{-1} (Avanzi, Gabellani, Delogu, Silvestro, Cremonese, et al. 2022). Solving this system of ordinary differential equations (ODEs), which is decremental to assess snowpack modifications, requires specific parametrizations for each of the mass fluxes involved.

Snowfall and rainfall mass fluxes are estimated from the total precipitation P through a phase partitioning scheme based on air temperature and relative humidity. S_f and R_f are thus computed as

$$\begin{cases} S_f = (1 - p_r) P \\ R_f = p_r P \end{cases}, \quad (1.8)$$

where p_r is the rainfall probability, defined as

$$p_r = \frac{1}{1 + e^{\alpha + \beta T_{air} + \gamma RH}}. \quad (1.9)$$

In the previous equation, T_{air} is the air temperature in $^{\circ}\text{C}$ and RH is the relative humidity in percent, while the fixed parameter $\alpha = 22$, $\beta = -2.7$ and $\gamma = -0.2$ are taken from Froidurot et al. 2014.

Snowmelt occurs only when both the current air temperature and its 10-day running mean exceed a threshold T_{τ} , set to be 1°C . This dual condition provides a simple way of accounting for the snowpack's cold content, defined as the energy required to warm a subfreezing snowpack until it reaches an isothermal 0°C state. By requiring sustained mild conditions, rather than a single warm hour, the model avoids triggering unphysical melt events that the snowpack's thermal inertia would not allow. Avanzi, Gabellani, Delogu, Silvestro, Cremonese, et al. 2022 estimate the melt rate through a hybrid temperature index and radiation driven melt parametrization. This formulation improves S3M's ability to reproduce the strong spatial variability that topography induces in snow cover extent and duration, since slope and aspect play a significant role in mountainous terrain that cannot be fully captured by air temperature alone. The hybrid approach adopted in S3M thus reads

$$M = m_{rad} \left[\frac{(1 - \alpha_S) S_r}{\rho_W \lambda_f} \right] \Delta t + c_M m_r (T_{air} - T_{\tau}), \quad (1.10)$$

where m_{rad} is a dimensionless coefficient that convert the potential radiative melt into actual melt, α_S is the snow albedo, S_r the incoming shortwave radiation, ρ_W the density of water, λ_f the specific latent heat of fusion and m_r the degree day parametes $[\text{mm } ^{\circ}\text{C}^{-1} \text{h}^{-1}]$. Both m_{rad} and m_r are calibration based.

In S3M, refreezing occurs whenever the air temperature drops below T_τ and is computed through a degree day approach as

$$R = -c_M m_r (T_{air} - T_\tau). \quad (1.11)$$

Since this expression depends solely on air temperature, refreezing is assumed to respond instantaneously to atmospheric cooling, without accounting for the internal structure or thermal gradient of the snowpack.

Ultimately, the runoff O is estimated by simulating liquid water movement through the snowpack using a matrix-flow approximation. This liquid flux is governed by physical properties of the snow cover, as

$$O = \alpha \frac{\rho_w g K_W}{\mu_W}, \quad (1.12)$$

where α is a conversion parameter, g the acceleration due to gravity, K_W the intrinsic permeability of water in snow and μ_W is water dynamic viscosity. This parametrization completes the system of ODEs that, together with those governing the evolution of snow density, are solved at each grid point using the forward Euler method. By default, S3M adopts an hourly integration time step, although this can be adjusted by the user to meet specific application needs.

1.3.2 Results

The IT-SNOW reanalysis provides daily, high-resolution estimates of snow properties, which are a key input for hydrological applications. Over the 2011-2025 period, peak SWE (intended as the total water volume stored in the snowpack) is on average $(11.9 \pm 5.2) \text{ Gm}^3$, with a minimum of 5.4 Gm^3 in 2022 and a maximum of 23.1 Gm^3 in 2014. This peak is typically reached on 4 March ± 17 days, with the earliest occurrence being 26 January 2023 and the latest 26 March 2013. This means that over the investigated period the maximum of SWE always precedes the 1st of April, which is the date commonly adopted in literature as a reference for peak SWE. The daily evolution of the total SWE volume, shown in Figure 1.6, exhibits great variability driven by thermal and meteorological conditions. The last few years stand out for a markedly reduced water volume stored in the Italian snowpack, with the partial exception of 2024, which nonetheless remained a slightly below average snow year. This is consistent with the persistent snow drought that characterised the early 2020s, driven by the combined effects of reduced precipitation and higher temperatures (Colombo et al. 2023).

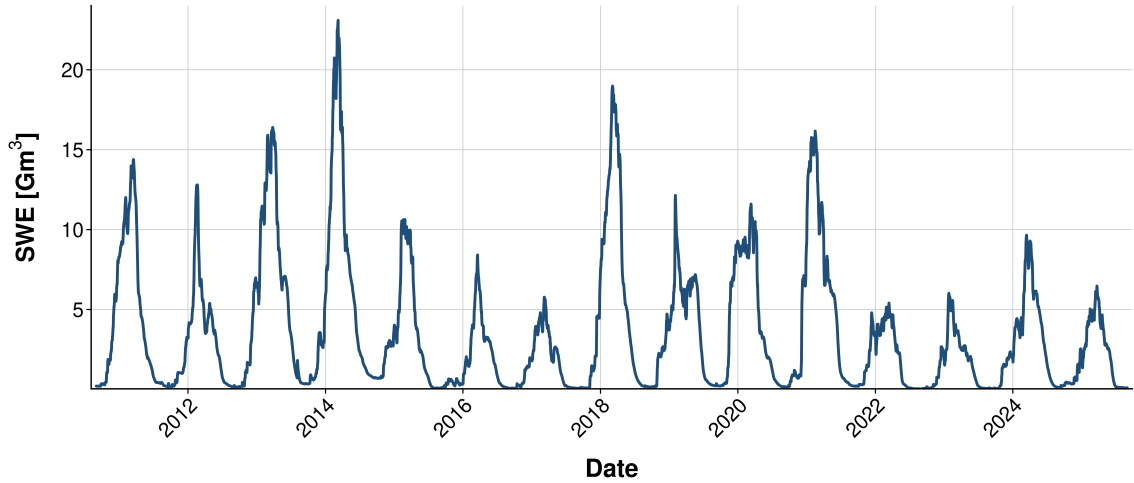


Figure 1.6: Daily evolution of the total water volume stored in the Italian snowpack, computed as snow water equivalent (SWE) integrated over the whole national domain, from 1 September 2010 to 31 August 2025. Data are from the IT-SNOW reanalysis (Avanzi, Gabellani, Delogu, Silvestro, Pignone, et al. [2023](#)).

Besides this strong temporal variability, the modelled snow cover also displays a marked spatial heterogeneity across Italy, with mountain regions experiencing longer and more intense snow seasons. Both panels of Figure 1.7 closely reproduce the orography of the main ranges, showing that the physically based S3M model captures the control exerted by topographic factors such as elevation and aspect on snowpack amount and persistence. According to IT-SNOW estimates, almost 87 % of the mean national winter SWE is stored within the five main Alpine river basins, namely Po River, Adige, Piave, Tagliamento and Brenta, whereas the Central Apennines account for about 7% of total SWE. The remaining fraction is distributed across the rest of the country (Avanzi, Gabellani, Delogu, Silvestro, Pignone, et al. [2023](#)).

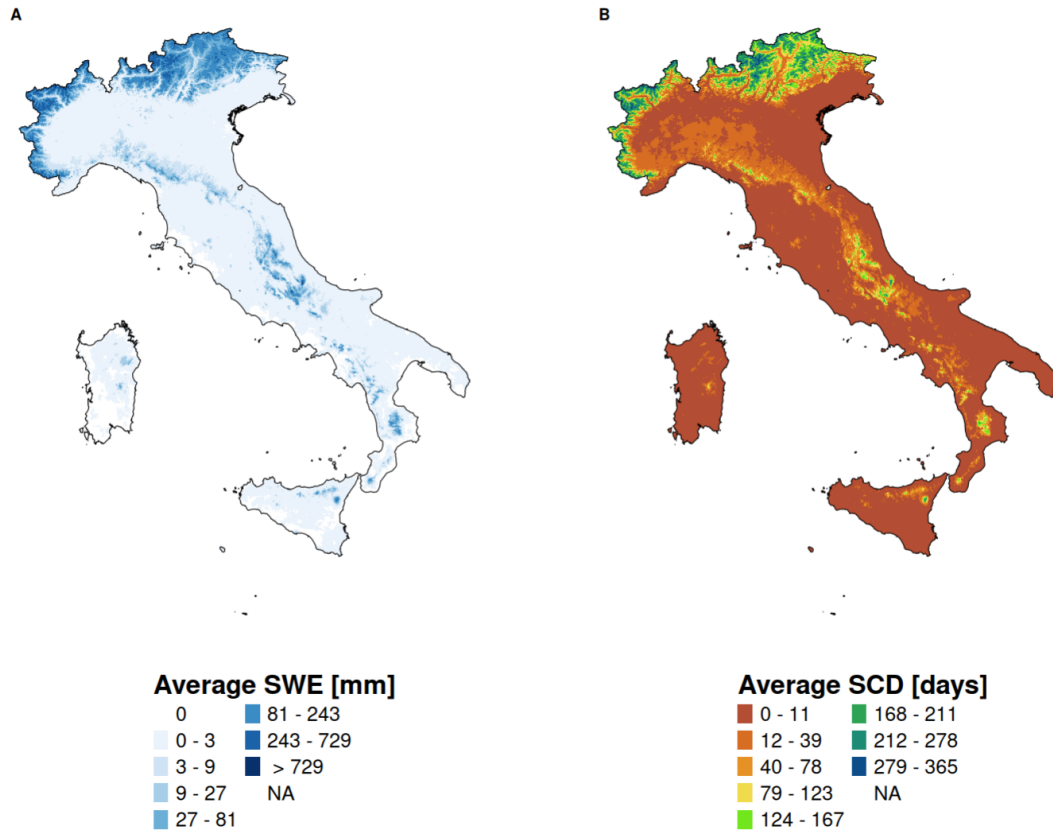


Figure 1.7: (a) Average seasonal snow water equivalent (SWE), computed as the mean of the November–June averages of each hydrological year over the period 1 September 2010–31 August 2025. (b) Average snow cover duration (SCD), computed as the mean of the annual SCD maps across the same period.

1.4 Po River Basin

This snow dataset focuses on the mountainous portion of the Po River District, an over-regional entity covering almost 87000 km^2 which comprises the Po River Basin together with some smaller adjacent catchments such as Conca and Reno. Assessing the amount of water stored as snow within this territory is decremental, as the Po River Basin is home to roughly one-third of Italy's total population and produces almost 55% of Italy's hydropower. This snow product is generated by an operational chain known as "J-snow", which is a hybrid approach that combines the physically based GEOtop model with in situ measurements of snow depth and snow cover extent. The dataset covers the hydrological years from 1992 to the 2021 at a daily time step, with a spatial resolution of approximately 500 m. Within each hydrological year, maps are available only between October 3rd and July 1st, a period that roughly corresponds to the snow season in non glacierized areas.

1.4.1 GEOtop

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1.4.2 Results

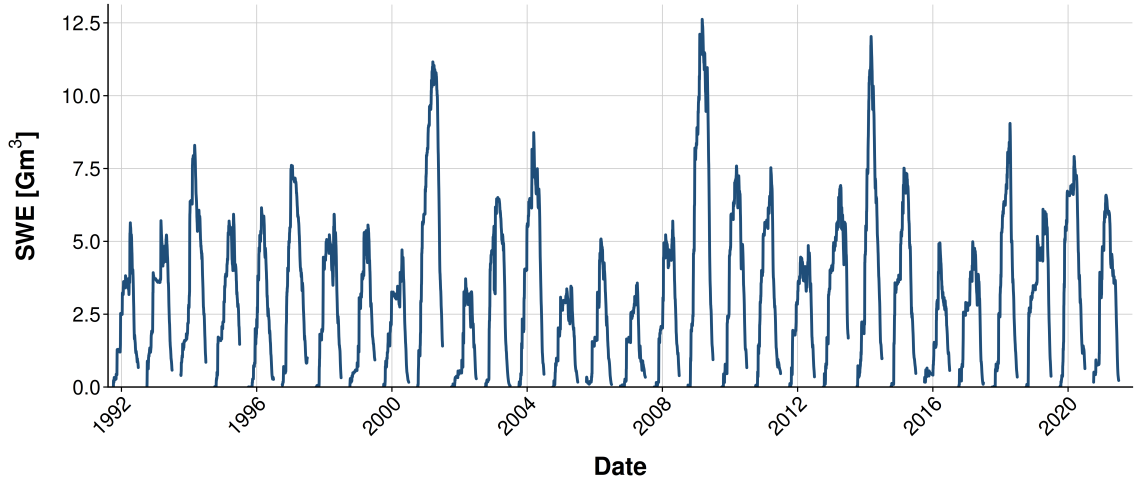


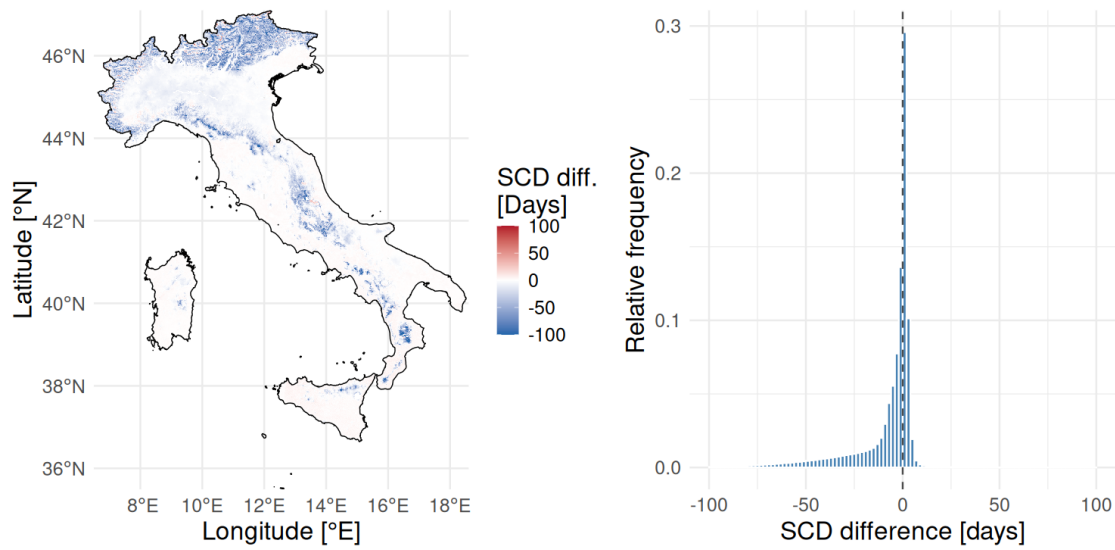
Figure 1.8: MODIS

1.5 Comparison

The snow datasets presented above differ from one another in several respects. ITSNOW and the Po Basin product are the outputs of two distinct physically based model, and both focus primarily on snow water equivalent, consistently with their hydrological purpose. ITSNOW, however, provides richer information than the latter, as it also includes maps of snow density, snow depth and liquid water content, which allow the user to gain a deeper insight into the state of the snowpack. MODIS, in contrast, describes snow cover alone through binary rasters. This discrepancy does not limit the possibility of a comparison, as SWE maps can be converted into binary rasters by means of (1.4) and (1.13). Since these datasets cover different domains, are provided at different spatial resolutions and in different coordinate reference systems, and span different periods, the comparisons reported hereafter are carried out over the largest common overlap, both in space and in time. For the sake of clarity, Table 1.2 summarizes the main characteristics of each snow product, thereby clarifying the rationale behind some of the comparison choices.

Table 1.2: Main characteristics of each snow product.

Dataset	Data type	Start date	End date	Study area	Spatial resolution	Reference system
MODIS	Snow cover	24-02-2000	30-09-2025	Italy	0.00417 °	WGS 84 (EPSG:4326)
ITSNOW	SWE	01-09-2010	31-08-2025	Italy	0.00506 °	WGS 84 (EPSG:4326)
Po Basin	SWE	03-10-1991	01-07-2021	Po Basin	500 m	ETRS89 (EPSG:3035)

**Figure 1.9:** MODIS

Appendix A

MODIS

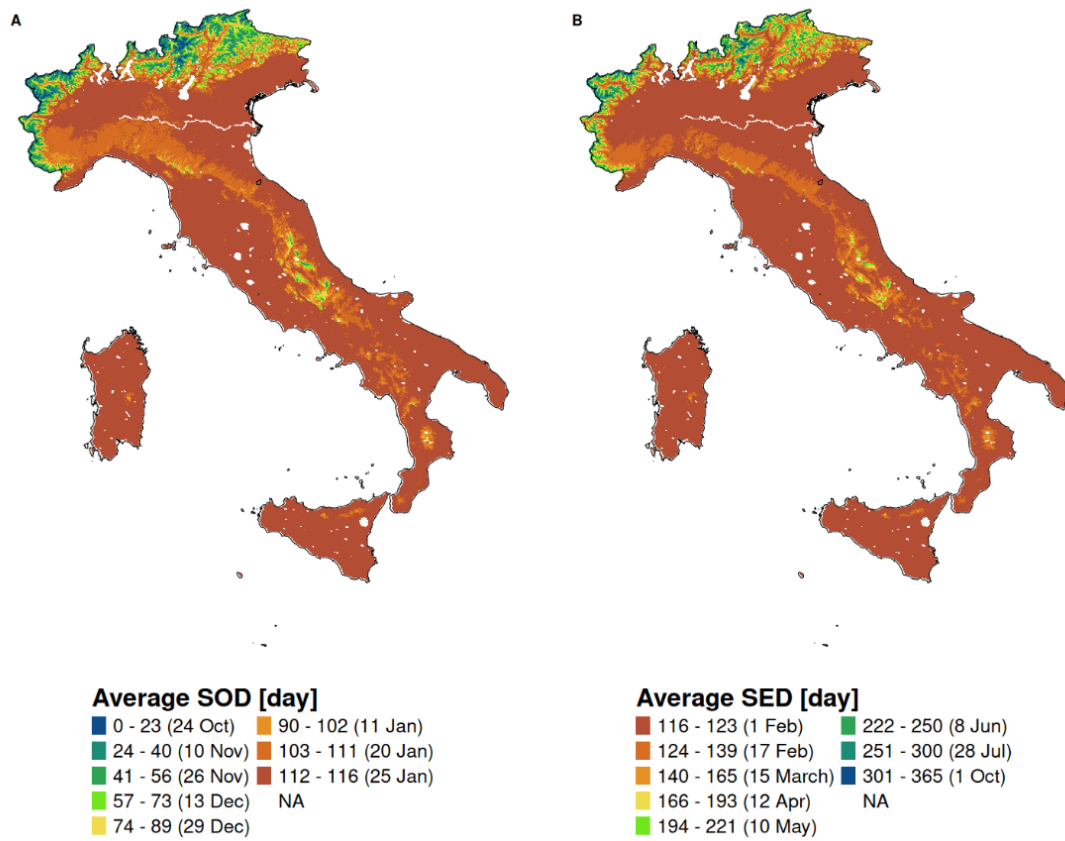


Figure A.1: Spatial distribution of the average Snow Onset Date (SOD, panel a) and Snow End Date (SED, panel b) across the Italian peninsula for the 2001-2025 period. As expected, mountain ranges are associated with an earlier onset and a later end of the snow season.

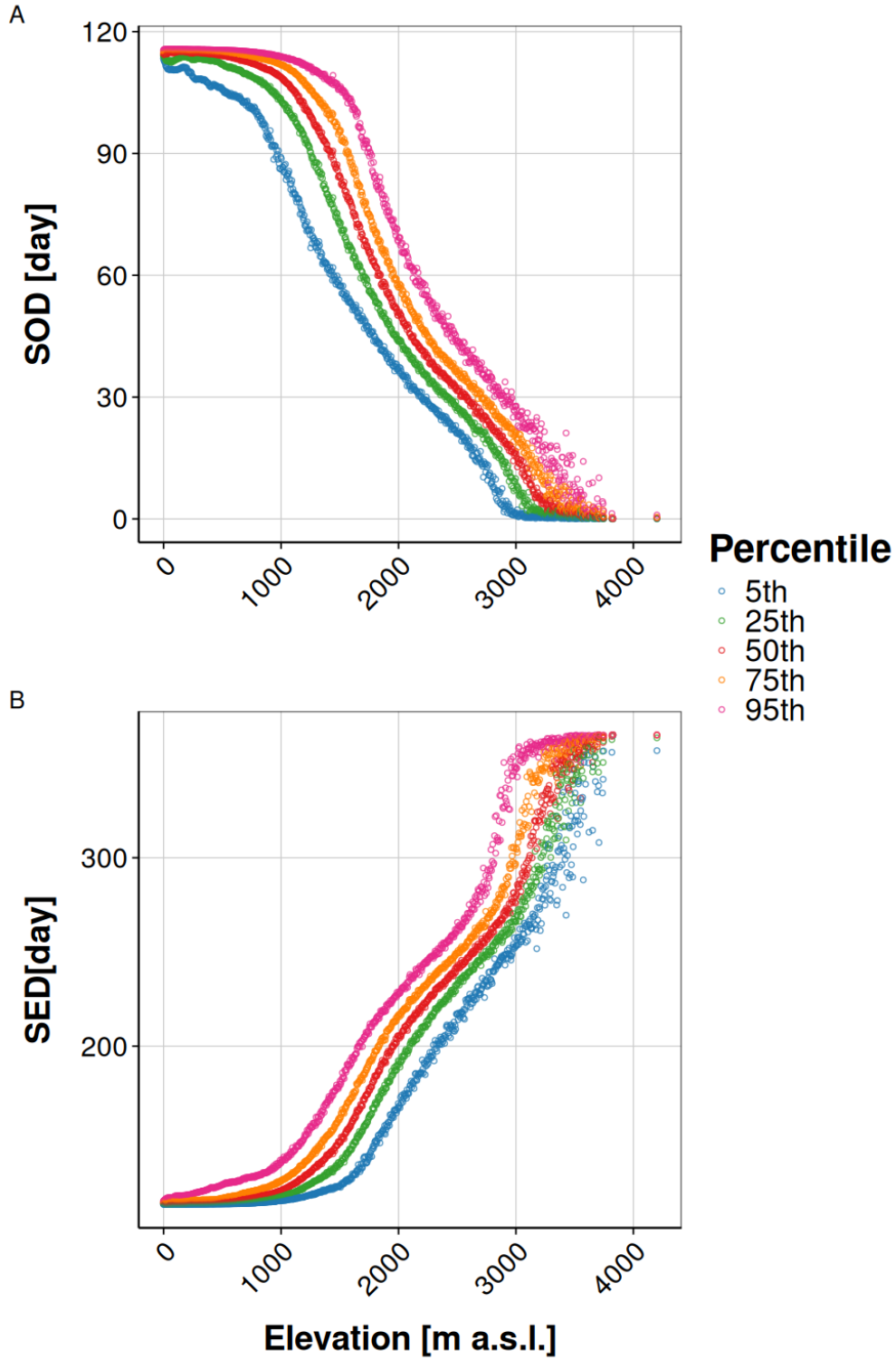


Figure A.2: Quantiles of the average Snow Onset Date (SOD, panel a) and Snow End Date (SED, panel b) distributions within each 5 m wide elevation band, for the 2001-2025 period. Both panels exhibit a monotonic trend with elevation, as high altitude alpine catchments experience earlier onset and later end of the snow season.

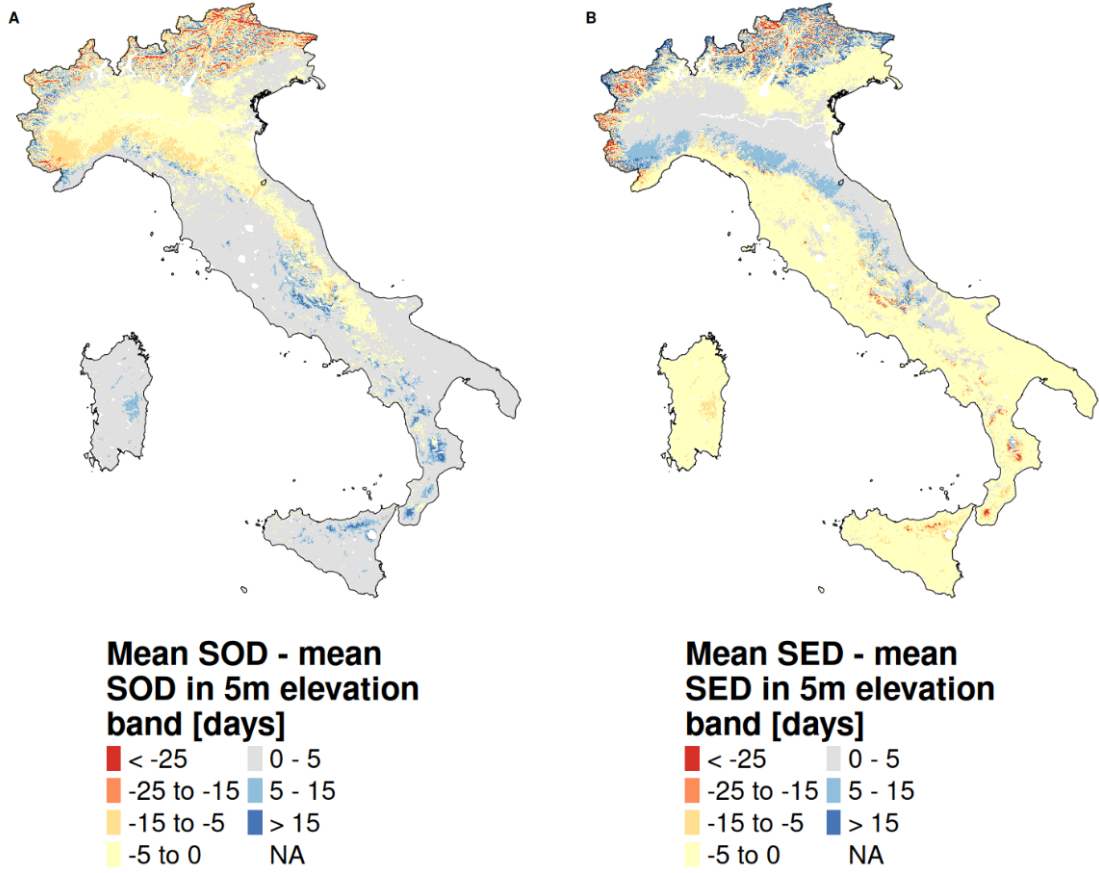


Figure A.3: (a) Spatial distribution of the difference between the SOD value of each individual pixel and the mean SOD of the corresponding 5 m wide elevation band. (b) Spatial distribution of the difference between the SED value of each individual pixel and the mean SED of the corresponding 5 m wide elevation band. The Eastern Apennines typically experience an earlier onset and a later end of the snow season relative to the elevation wise Italian average. This behaviour could be related to cold air masses originating from the Balkans. In the Alps, pixel aspect plays a significant role, as south facing slopes often experience earlier melt.

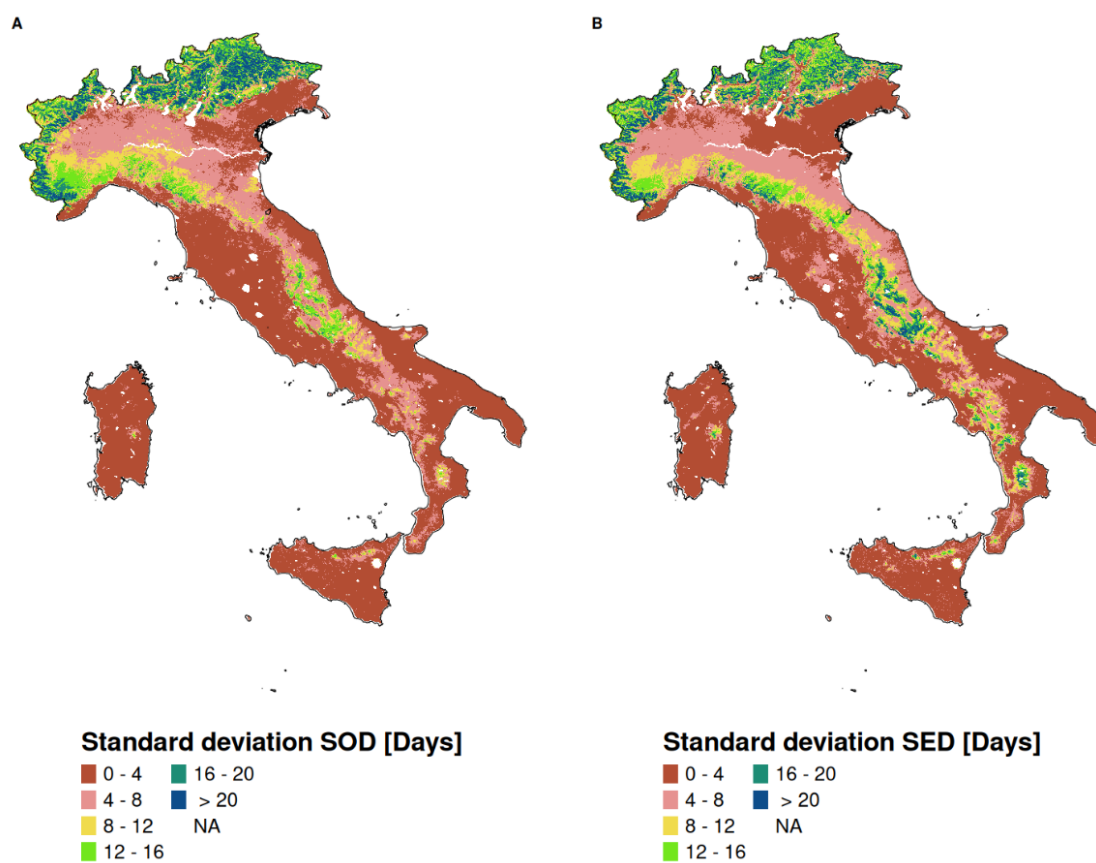


Figure A.4: Spatial distribution of Snow Onset Date (SOD, panel a) and Snow End Date (SED, panel b) standard deviation from the average value computed over the 2001-2025 period.

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